6. The polynomial $x^3 - kx^2 + 9x$ has three real zeros if and only if, k satisfies

(1) $|k| \leq 6$ (2) $k = 0$ (3) $|k| > 6$ (4) $|k| \geq 6$

7. The number of real numbers in $[0, 2\pi]$ satisfying $\sin^4 x - 2\sin^2 x + 1$ is

(1) 2 (2) 4 (3) 1 (4) ∞

8. If $x^3 + 12x^2 + 10ax + 1999$ definitely has a positive zero, if and only if

(1) $a \geq 0$ (2) $a > 0$ (3) $a < 0$ (4) $a \leq 0$

9. The polynomial $x^3 + 2x + 3$ has

(1) one negative and two imaginary zeros (2) one positive and two imaginary zeros

(3) three real zeros (4) no zeros

10. The number of positive zeros of the polynomial $\sum_{j=0}^{n} {}^nC_r(-1)^r x^r$ is

(1) 0 (2) n (3) $< n$ (4) r

SUMMARY

In this chapter we studied

- Vieta's Formula for polynomial equations of degree 2,3, and n>3.
- **The Fundamental Theorem of Algebra** : A polynomial of degree $n \geq 1$ has at least one root in $\mathbb{C}$.
- **Complex Conjugate Root Theorem** : Imaginary (nonreal complex) roots occur as conjugate pairs, if the coefficients of the polynomial are real.
- **Rational Root Theorem** : Let $a_n x^n + \cdots + a_1 x + a_0$ with $a_n \neq 0$ and $a_0 \neq 0$, be a polynomial with integer coefficients. If $\frac{p}{q}$, with $(p, q) = 1$, is a root of the polynomial, then p is a factor of a_0 and q is a factor of a_n.
- Methods to solve some special types of polynomial equations like polynomials having only even powers, partly factored polynomials, polynomials with sum of the coefficients is zero, reciprocal equations.
- **Descartes Rule :** If p is the number of positive roots of a polynomial $P(x)$ and s is the number of sign changes in coefficients of $P(x)$, then $s - p$ is a nonnegative even integer.

ICT CORNER

https://ggbm.at/vchq92pg or Scan the QR Code

Open the Browser, type the URL Link given below (or) Scan the QR code. GeoGebra work book named "12th Standard Mathematics" will open. In the left side of the work book there are many chapters related to your text book. Click on the chapter named "Theory of Equations". You can see several work sheets related to the chapter. Select the work sheet "Relation between roots and co-efficients"

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